

Continuous Random Variables



Verifying a valid probability density function

- 1 A continuous random variable X has probability density function $f(x) = \frac{1}{4}$ for $0 \leq x \leq 4$, and $f(x) = 0$ otherwise. Verify that this is a valid pdf.
 - 2 Determine the value of c such that $f(x) = c$ for $2 \leq x \leq 6$, and $f(x) = 0$ otherwise, is a valid pdf.
-

Finding an unknown constant in a pdf

- 3 A continuous random variable X has pdf $f(x) = kx$ for $0 \leq x \leq 5$, and $f(x) = 0$ otherwise. Find k .
 - 4 A continuous random variable X has pdf $f(x) = k(4 - x)$ for $0 \leq x \leq 4$, and $f(x) = 0$ otherwise. Find k .
 - 5 A continuous random variable X has pdf $f(x) = kx^2$ for $0 \leq x \leq 3$, and $f(x) = 0$ otherwise. Find k .
-

Finding probabilities by integrating the pdf

- 6 Using $f(x) = \frac{2}{25}x$ for $0 \leq x \leq 5$, find $P(X \leq 2)$.
 - 7 Using the same pdf, find $P(2 \leq X \leq 4)$.
 - 8 Using $f(x) = \frac{1}{8}(4 - x)$ for $0 \leq x \leq 4$, find $P(X \geq 2)$.
 - 9 Using $f(x) = \frac{1}{9}x^2$ for $0 \leq x \leq 3$, find $P(X \leq 1)$.
-

The cumulative distribution function

- 10 For the pdf $f(x) = \frac{2}{25}x$ on $0 \leq x \leq 5$, find the cumulative distribution function $F(x)$.
 - 11 Using $F(x) = \frac{x^2}{25}$, find $P(X \leq 3)$ and hence $P(X > 3)$.
 - 12 For the pdf $f(x) = \frac{1}{8}(4 - x)$ on $0 \leq x \leq 4$, find the cumulative distribution function $F(x)$.
 - 13 For the pdf $f(x) = \frac{1}{9}x^2$ on $0 \leq x \leq 3$, find the cumulative distribution function $F(x)$.
-

Using the cdf to find probabilities

- 14 Given $F(x) = \frac{x^2}{25}$ for $0 \leq x \leq 5$, find $P(1 \leq X \leq 4)$.
- 15 Using $F(x) = \frac{x}{2} - \frac{x^2}{16}$, find $P(1 \leq X \leq 3)$.
- 16 Using $F(x) = \frac{x^3}{27}$ for $0 \leq x \leq 3$, find $P(1 \leq X \leq 2)$.
-

Expectation of a continuous random variable

- 17 Using $f(x) = \frac{2}{25}x$ for $0 \leq x \leq 5$, find $E(X)$, using $E(X) = \int xf(x) dx$.
- 18 Using the uniform pdf $f(x) = \frac{1}{4}$ for $0 \leq x \leq 4$, find $E(X)$, and verify it matches the midpoint of the interval.
- 19 Using $f(x) = \frac{1}{9}x^2$ on $[0, 3]$, find $E(X)$.
-

Variance of a continuous random variable

- 20 Using $f(x) = \frac{2}{25}x$ on $[0, 5]$ and $E(X) = \frac{10}{3}$ (found earlier), find $\text{Var}(X)$, using $\text{Var}(X) = E(X^2) - [E(X)]^2$.
- 21 Using the uniform pdf $f(x) = \frac{1}{4}$ on $[0, 4]$ and $E(X) = 2$ (found earlier), find $\text{Var}(X)$.
- 22 Using $f(x) = \frac{1}{9}x^2$ on $[0, 3]$ and $E(X) = \frac{9}{4}$ (found earlier), find $\text{Var}(X)$.
-

Median and quartiles

- 23 Using the cdf $F(x) = \frac{x^2}{25}$ for $0 \leq x \leq 5$, find the median m of X , solving $F(m) = 0.5$.
- 24 Using the same cdf, find the lower quartile Q_1 , solving $F(Q_1) = 0.25$.
- 25 Using $F(x) = \frac{x}{2} - \frac{x^2}{16}$ for $0 \leq x \leq 4$, find the upper quartile Q_3 , solving $F(Q_3) = 0.75$.
- 26 Using the cdf $F(x) = \frac{x^3}{27}$ for $0 \leq x \leq 3$, find the median m of X , to three decimal places.
-

A well-designed word problem: waiting time at a bus stop

- 27 This question has four parts. The waiting time X (minutes) for a bus is modeled by $f(x) = \frac{3}{64}x^2$ for $0 \leq x \leq 4$, and $f(x) = 0$ otherwise. (a) Verify this is a valid pdf. (b) Find the cdf $F(x)$, and use it to find the probability a passenger waits more than 3 minutes. (c) Find $E(X)$, the mean waiting time. (d) Find the median waiting time, to three decimal places.